

5. Perform the following tasks using Git-Hub Desktop
- Initialize a Git repository for a data science project
 - Push it to GitHub to enable version control
 - Create a new branch named GIT-EXAM-1 and create new file named as program2 and write a program to check if given string is palindrome
 - Create a python file name as program1.py and write a python program to print squares of numbers from 1 to 10
 - Commit the code and push the changes and check the changes on remote repository (GitHub)
 - Create a directory named as IGNORE and add test1.txt, test2.txt files into it
 - Create .gitignore file and add the IGNORE directory into it to ignore the changes made to it
 - Commit and push the code and demonstrate that IGNORE directory and its files are not getting pushed on remote

✅ **Step 1: Initialize Repository (GitHub Desktop)**

- Open **GitHub Desktop**
- Click **File → New Repository**
- Fill details:
 - Name: data-science-project
 - Local path: choose folder
- Click **Create Repository**

✅ **Step 2: Publish to GitHub**

- Click **Publish Repository** (top right)
- Choose: Keep it Public/Private
- Click **Publish**

✅ **Step 3: Create Branch GIT-EXAM-1**

- Click **Current Branch** (top bar)
- Click **New Branch**
- Name: GIT-EXAM-1 Click **Create Branch**

✅ **Step 4: Create program2.py (Palindrome)**

- Go to your project folder
- Create file: program2.py

👉 Add code:

Palindrome check

```
string = input("Enter string: ")
if string == string[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

✅ **Step 5: Create program1.py (Squares)**

Create file: program1.py

Squares from 1 to 10

```
for i in range(1, 11):
    print(i*i)
```

✅ **Step 6: Commit & Push**

- Open **GitHub Desktop**
- You will see changes listed
- Add message:
Added program1 and program2
- Click **Commit to GIT-EXAM-1**

5. Click **Push origin** 👉 ✓ Check GitHub → files visible

✅ **Step 7: Create IGNORE Folder**

1. Inside project folder:
 - Create folder: IGNORE
 - Create files:
 - test1.txt
 - test2.txt

✅ **Step 8: Create .gitignore (Create outside ignore folder in datasci folder)**

1. Create file: .gitignore
2. Add: IGNORE/

✅ **Step 9: Commit .gitignore**

1. Go to GitHub Desktop
2. You will see .gitignore change
3. Add message:
Added gitignore to ignore IGNORE folder
4. Click **Commit to GIT-EXAM-1**
5. Click **Push origin**

⚠ **IMPORTANT (If IGNORE files appear):**

Open data-science-project folder

- git branch You should see * GIT-EXAM-1

✅ **Now run the commands here**

git rm -r --cached IGNORE

git commit -m "Removed IGNORE folder from tracking"

git push origin GIT-EXAM-1 **AGAIN FOLLOW STEP 7,8,9**

✅ **Step 10: Verify (VERY IMPORTANT)**

👉 On GitHub:

- ✓ program1.py → visible
- ✓ program2.py → visible
- ✓ .gitignore → visible
- ✗ IGNORE/ → NOT visible

🔥 **Viva Line** 👉 You can say:

Used .gitignore to prevent the IGNORE directory from being pushed to GitHub.